

BRENT A STATION 63113, United KingdomWMO: **996070**Lat: **61.000N**Lon: **1.708E**Elev: **98**StdP: **14.64**Time Zone: **0.00 (EUW)**Period: **02-15**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	32.1	33.7	N/A	N/A	N/A	N/A	N/A	N/A	49.3	38.2	45.8	39.4	21.4	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.0	66.0	N/A	63.3	N/A	61.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	12.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 44.0	39.0	34.4	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.9	41.9	41.4	42.1	44.5	47.4	52.0	55.9	57.4	54.2	49.1	46.3	42.6		
	DBStd	6.54	3.08	3.66	3.65	3.46	3.80	3.75	3.56	3.29	3.10	3.44	3.19	4.05		
	HDD50	1419	250	240	246	167	98	16	0	0	4	55	114	229		
	HDD65	6235	715	660	711	616	546	390	285	239	324	493	562	694		
	CDD50	665	0	0	0	1	17	77	182	228	130	28	2	0		
	CDD65	4	0	0	0	0	0	0	2	2	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	17.8	22.0	19.9	19.6	16.4	15.6	13.4	14.8	14.1	17.3	18.4	21.3	21.1		
(15) Precipitation	PrecAvg	52.8	6.3	5.2	5.0	2.8	1.9	2.6	3.2	3.9	4.6	5.6	5.5	6.1		
	PrecMax	65.0	9.9	10.1	9.3	4.8	3.3	5.2	6.0	6.4	7.5	8.5	8.7	9.4		
	PrecMin	37.6	3.0	2.3	1.8	1.0	0.9	1.0	1.6	2.6	1.9	2.3	1.7	2.6		
	PrecStd	6.5	1.9	2.1	1.8	0.9	0.6	0.9	0.9	1.0	1.3	1.6	1.5	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.2	48.9	50.3	54.3	60.3	67.1	69.2	70.2	64.6	58.1	53.0	51.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	48.0	47.7	48.4	51.8	56.1	62.5	66.1	65.7	61.4	55.9	51.8	49.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	47.1	46.9	47.3	50.0	54.0	58.9	62.7	63.5	59.3	54.4	50.9	48.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	46.3	46.2	46.4	48.5	52.0	56.3	60.6	61.6	58.2	53.3	50.2	48.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	4.6	4.7	4.5	4.2	4.0	4.0	4.0	3.5	3.8	4.6	5.1		
		MCDBR	5.2	5.0	4.9	6.4	7.0	7.6	7.5	6.9	4.7	4.0	5.2	5.7		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.334	0.355	0.368	0.380	0.370	0.362	0.380	0.370	0.364	0.346	0.321	0.388		
	taud		2.067	2.191	2.271	2.307	2.409	2.471	2.437	2.463	2.432	2.311	2.133	2.322		
	Ebn at Noon		111	190	235	257	271	275	266	259	238	193	116	71		
	Edh at Noon		14	23	29	34	33	31	32	29	25	20	13	9		
(44) All-Sky Solar Radiation	RadAvg		62	220	580	1129	1574	1605	1456	1159	704	331	97	39		
	RadStd		8	28	64	94	110	173	129	105	55	40	12	3		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)		N/A	+1.17	+1.12	N/A	N/A	N/A	-184	-241	N/A	N/A	N/A

Nomenclature: See separate page